

AWS → CLOUDFLARE MIGRATION · PROJECT DOCUMENTATION

WAF Rule Mapping Document

Supporting Document 2 - AWS WAF → Cloudflare Rule Parity

SaaS Edge Layer - AWS CloudFront/WAF/Shield → Cloudflare

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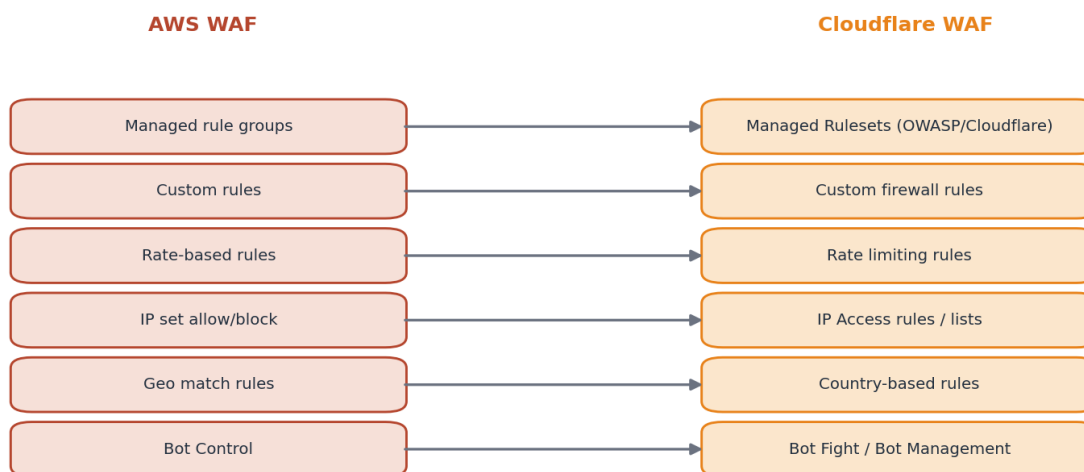
1. Overview & Approach

This document maps every category of AWS WAF rule to its Cloudflare equivalent. Getting this mapping right was the most important part of the migration, WAF rule parity is where migrations like this go wrong if they are rushed. The goal was to recreate the same protection in Cloudflare, verify it, and only then move traffic.

1.1 Approach

1. Inventory every existing AWS WAF rule and group across the web ACL.
2. Identify the Cloudflare feature that provides the equivalent behavior.
3. Recreate each rule in Cloudflare with matching conditions and actions.
4. Test each rule on a sandbox before any production traffic moved.
5. Confirm parity by reviewing WAF events after cutover.

WAF Rule Mapping — AWS WAF → Cloudflare



Every existing rule was recreated and verified for parity — the step where rushed migrations fail.

Figure 1 - AWS WAF rule categories mapped to Cloudflare

Guiding principle

Match behavior, not just names. Each rule was recreated so it produces the same allow, block, or challenge outcome in Cloudflare as it did in AWS WAF then verified rather than assumed.

2. AWS Managed Rule Groups → Cloudflare Equivalents

AWS WAF managed rule groups provide pre-built protections (such as common web exploits). In Cloudflare these are covered by the managed rulesets, which include OWASP-style and Cloudflare-maintained protections.

AWS WAF	Cloudflare equivalent	Notes
Core / common rule sets	Cloudflare Managed Ruleset	Baseline protection against common web attacks.
Known bad inputs	Cloudflare Managed Ruleset	Blocks known malicious request patterns.
OWASP-style groups	OWASP Core Ruleset	Tunable sensitivity / paranoia level.

Managed rulesets are maintained by the provider, so they stay current without manual upkeep - the same benefit AWS managed rule groups offered.

3. Custom Rule Migration

Custom AWS WAF rules, application-specific conditions written for this client were recreated as Cloudflare custom firewall rules. Each condition (matching on URI, headers, methods, or query strings) was rebuilt with the same logic and action.

AWS WAF concept	Cloudflare equivalent
Custom rule with match conditions	Custom firewall rule with expression
Match on URI / path	Expression on <code>http.request.uri.path</code>
Match on headers / method	Expression on <code>http.request.headers / method</code>
Allow / Block / Count action	Allow / Block / Log action

4. Rate Limiting Rules

AWS WAF rate-based rules were recreated as Cloudflare rate limiting rules. These protect against abuse by limiting how many requests a single client can make in a given window.

Element	AWS WAF	Cloudflare
Rule type	Rate-based rule	Rate limiting rule
Threshold	Requests per 5-minute window	Requests per configurable window
Scope	By IP / matching criteria	By IP / path / matching criteria
Action	Block when exceeded	Block or challenge when exceeded

Tuning note

Thresholds were set to match the existing AWS behavior, then reviewed against real traffic so legitimate users were not affected while abuse was still blocked.

5. IP Allow / Block Lists

AWS WAF IP sets (allow and block lists) were recreated using Cloudflare IP Access rules. The same addresses and ranges were carried over with the same intent.

Purpose	AWS WAF	Cloudflare
Block list	IP set referenced in a block rule	IP Access rule - Block
Allow list	IP set referenced in an allow rule	IP Access rule - Allow
Ranges (CIDR)	CIDR entries in the IP set	CIDR entries in the access rule

6. Country-Based Rules

AWS WAF geo-match rules were recreated as Cloudflare country-based rules. These allow or block traffic based on the country a request originates from.

Element	AWS WAF	Cloudflare
Match	Geo match condition (country)	Expression on ip.geoip.country
Action	Allow / Block by country	Allow / Block / Challenge by country

7. Bot Protection Rules

AWS WAF Bot Control was mapped to Cloudflare's bot protection. Cloudflare provides bot mitigation that distinguishes good bots (like search engines) from harmful automated traffic.

AWS WAF	Cloudflare equivalent	Notes
Bot Control rule group	Bot Fight Mode / Bot Management	Challenges or blocks suspicious automated traffic.
Allow verified bots	Verified bot allowance	Legitimate crawlers continue to work.

8. Testing & Validation

Once every rule category was mapped and recreated, the full ruleset was validated on a sandbox before production traffic moved, and then monitored after cutover.

8.1 Validation steps

1. Confirm each managed and custom rule is present and active in Cloudflare.
2. Send test requests that should be blocked and confirm they are blocked.
3. Send legitimate requests and confirm they pass without disruption.
4. Verify rate limiting triggers at the expected thresholds.
5. Review Cloudflare WAF events after cutover to confirm rules fire as expected.

8.2 Parity checklist

Rule category	Recreated	Validated
Managed rule groups	Yes	Yes
Custom rules	Yes	Yes
Rate limiting	Yes	Yes
IP allow / block lists	Yes	Yes
Country-based rules	Yes	Yes
Bot protection	Yes	Yes

Outcome

Every category of AWS WAF protection was recreated in Cloudflare and verified before and after cutover, so the client moved to the new edge without losing any of the protection they had.